

CHE 301/341/433 Fall 2019 and Spring 2020 Self-Study Report

Instructor: Michael Caracotsios

Course Comments / Plans

For the past five years that I have been teaching at the Chemical Engineering Department at UIC, my evaluations have been fairly consistent mostly in the range 4-5 with an overall score of Excellent. The principal object or goal of my teaching method is to find a point of view from which the subject appears in its greatest simplicity. I strongly believe that the ultimate goal of teaching is to enhance learning without focusing on the testing aspect which tends to cause a lot of anxiety and uneasiness in students. As such, without sacrificing the fundamental aspects of my courses (Thermodynamics, Process Control and Simulation) I design problems not from available textbooks but from my own personal experience of over 30 years in the chemical, petrochemical and refining industries. Although these problems are extremely challenging and sometimes with no obvious solutions, the students tend to enjoy them enormously and they always give positive feedback in their evaluations. (although they do complain about the overwhelming load)

One aspect of my teaching that seems to upset some (but not all) students is that I do not follow a traditional textbook. I recommend a number of textbooks in the beginning of each semester but I do not endorse or adopt any of them. It find it incredulous that I would have to exclusively follow someone else's idea on a particular engineering subject and especially Thermodynamics and Process Control. So I write my own notes from a variety of sources and my personal experience in industry and distribute them to the students for free. Most students enjoy the notes and give positive feedback but there is always a small number that prefers the traditional way of teaching. Furthermore I find it unconscious that the textbooks are so expensive and I don't expect my students to have the means to pay for these textbooks. As a result I will continue on my method of writing and improving and distributing my own notes. Hopefully in a short time I will be able to combine all my notes in a manuscript that I can distribute for free.

Another aspect of my teaching that seems to annoy some students is that I do not post solutions. I explain to them that the reason for that is, if I post the solutions most of the students copy them or take pictures with their smart devices and end up learning absolutely nothing. Instead I encourage the students, if they don't understand parts of the assignment to come to me (my door is always open) and we can have a conversation in order to enhance their learning.

Finally another major aspect from the evaluations appears to be my stringent requirements for attendance, tardiness and work ethics. Simply I cannot tolerate laziness and lack of respect from those who walk in late interrupting my lectures as they navigate through occupied seats to find an empty one. Attendance is mandatory in my lectures, late students are not allowed in the classroom and outstanding work ethic is rewarded handsomely in terms of improving the grades when I finally assign them. Some students seem to have a problem with that, but overall most of them support my methods.

To close up my self-study report I would like to stress again that my methods of teaching are designed to enhance learning without placing an unfair burden on the students through testing.